Subsurface to interface: 3D geological modelling and interactive web visualisation of natural hydrogen

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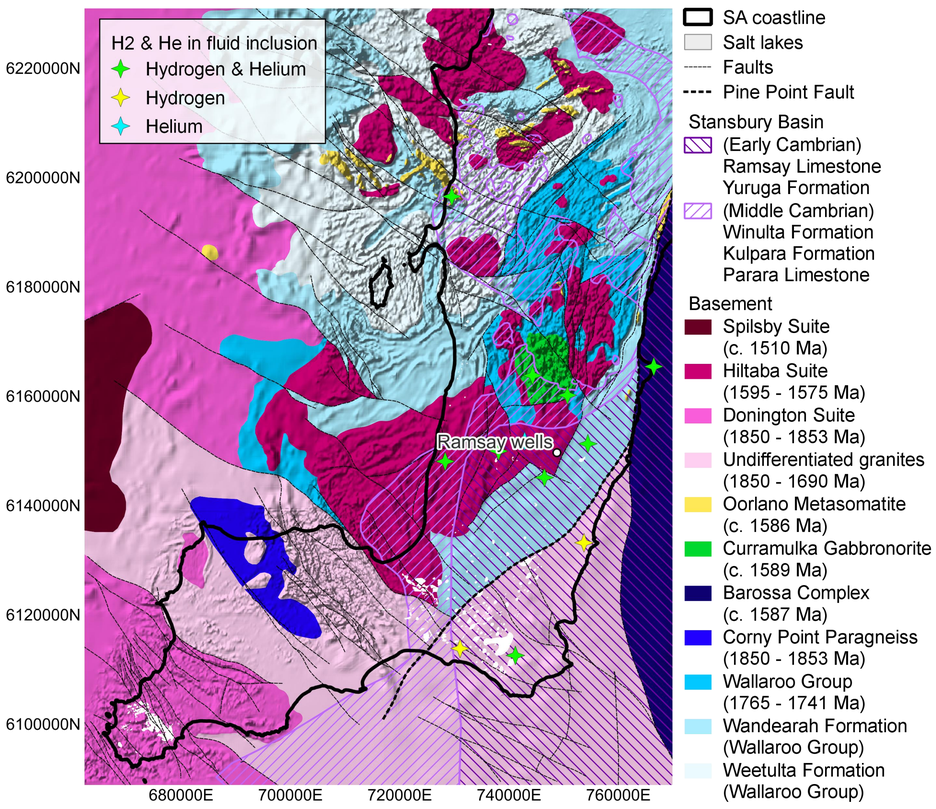
The forecasted demand for low-carbon emission hydrogen and the commercial operation of natural hydrogen extraction in Mali since 2012 have prompted governments to establish legal frameworks for natural hydrogen. Tasmania, SA and NT have legislated natural hydrogen exploration.

Case Study: Yorke Peninsula

In late 2023, **Ramsay 1** and **Ramsay 2** were drilled into Yorke Peninsula, South Australia by Gold Hydrogen Limited with the primary objective of validating anomalously high natural hydrogen levels, previously measured in the 1930s. The were validated, containing naturally occurring raw gas levels for hydrogen (H_2) of up to 86% and helium (He) up to 6.8% (non-air corrected).[3]



Quaternary Cover	Unconsolidated sediments	No gas mention.
Ramsay Limestone	Fossiliferous limestone with basal sandy ooids, transitioning upward to black nodular lime mudstone.	No gas mention.
Minlaton Formation	Grey-green shale to red-brown silty layers with conglomerates, interbedded sandstone, and stromatolitic dolostone.	No gas mention but often recognised as a potential seal for hydrocarbons because of shale layers.
Parara Limestone	transgressive marine unit composed of argillaceous, nodular limestone, with stromatolitic and tuffaceous interbeds. Deposited under oscillating aerobic-anaerobic conditions and reflects a deepening environment with phosphate coatings forming in condensed sections.	Hydrogen purity peaked at **95.8% at 531m.** **Significant permeability from fractures**.
Kulpara Formation	Interweaving limestone and dolomite, with stromatolitic and oolitic grainstones frequently dolomitised, enhancing porosity and vuggy textures. Peritidal carbonate succession deposited on a carbonate shelf from the Gawler Craton, reflects tidal influences and transgressive-regressive cycles.	**180m thick helium play zone** identified in a dolomitic section between 612m and 778m in the Ramsay 2 well. Air-corrected **helium concentration up to 17.5% at 778m**. **Significant permeability from fractures** and continuous flow to the surface during shut-in periods.
Winulta Formation	Shallow marine siltstone and calcareous sandstone, featuring bioturbation and Tommotian fossils, deposited in subtidal environments.	No gas mention.
Weathered granite / **Crystalline Granite**	Granite underlying all sedimentary units with fractures and weathering that enhance porosity and permeability, creating fluid pathways.	Ramsay 2 revealed **helium purity between 20-25%** in basement-derived gas. Helium from radioactive decay within granite fractures.

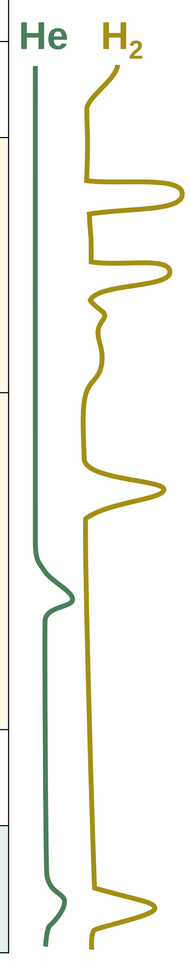
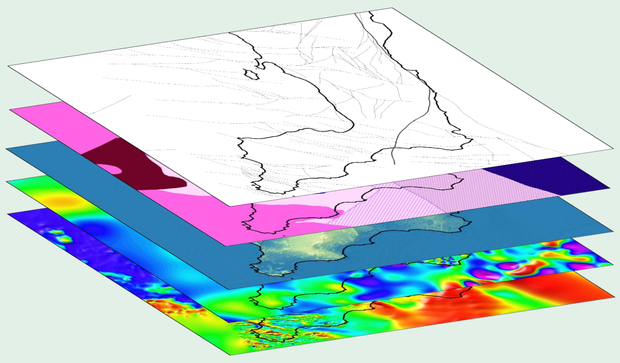


Figure 1 (top): Summary of formations encountered in Ramsay 1 and 2 wells, detailing geological descriptions, gas relevance and approximate comparative concentration of helium and hydrogen along the stratigraphic column.[1,3]
Figure 2 (left): Map of Yorke Peninsula highlighting gas occurrences in fluid inclusions[3], fault lines and geological units.[4]

3D Geological Modelling

Integrates geophysical, lithological, and structural data from regional surveys[4], historical drill holes[4], and recent wellbores from the Ramsey Project. By interpreting this dataset and applying geostatistical methods, a 3D litho-structural model is created using Rock Flow Dynamics' tNavigator software.

Well	Easting	Northing	Well	Easting	Northing
Minlaton 1	746460.2	6145064.4	CUR D 1B	751769.9	6150872.5
Minlaton 2	748883.7	6151167.2	Ramsay Oil Bore	748185.9	6149653.42
SYC 101	742866.3	6151721.8			



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Lithology interpolations

- **Kriging:** Smooth, conditional simulation of lithological contacts.
- **Natural Neighbour:** Preserves local accuracy and sharp changes, ideal for irregular well spacing.
- **Other interpolation methods** are applied as appropriate based on data format.

Fault plane characterisation

- Using averaged dips of identified fractures if data is available, or
- Approximated according to local geological structures.

These methods ensure geological models are accurate and align with local subsurface conditions.

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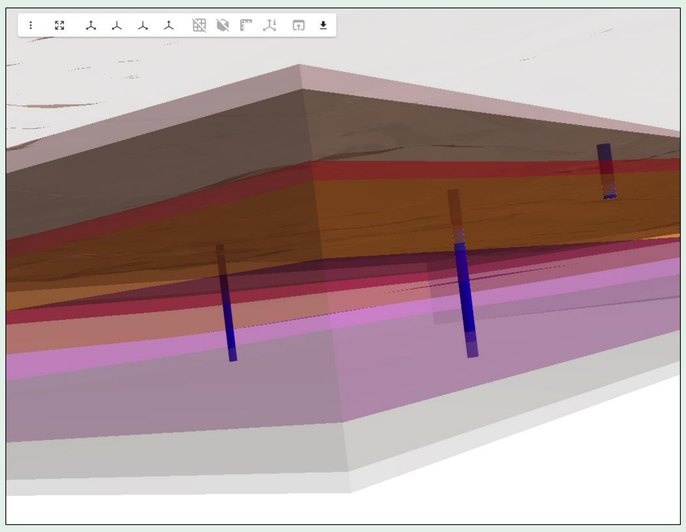


Model Validation

The model undergoes a comparative analysis with surface soil gas survey data and/or seismic data, which independently validates the identified locations of natural H2 migration pathways and traps in the model.

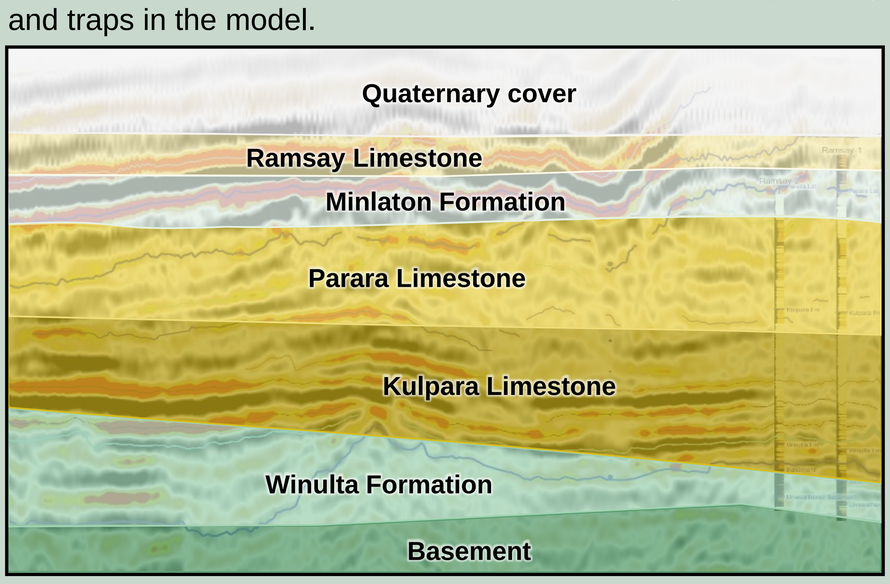
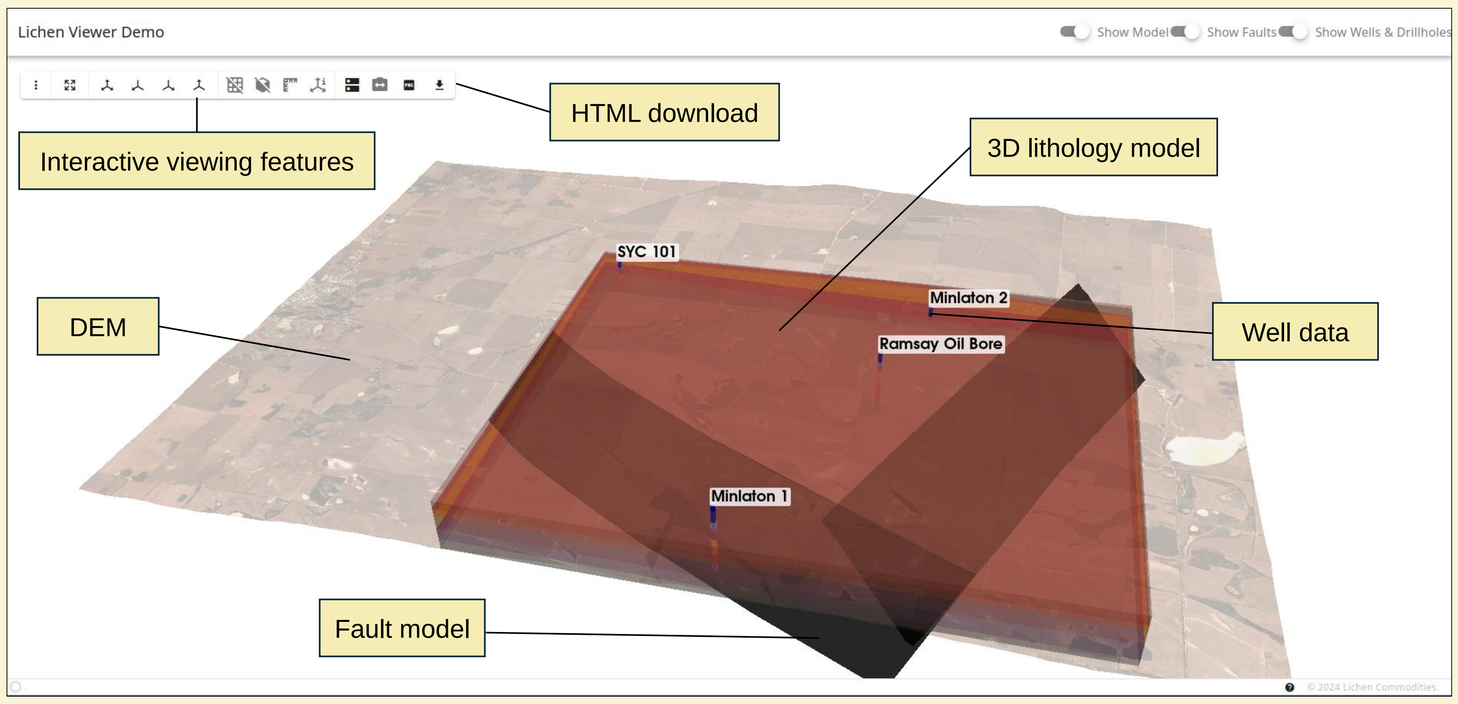


Figure 3: Model validation process comparing the boundaries of the generated 3D model with seismic data from line Y82-A1.[2] Note that lithological data from Ramsay 1 and 2 are not included in the model, which used only five legacy drillholes as outlined in the previous section. However, the resulting surfaces aligned satisfactorily on a general scale with the seismic interpretation from Ramsay 1 and 2.

Interactive Web Visualisation

Development of an interactive web browser-hosted using the Python library PyVista, for resource projects, enabling stakeholders to gain an intuitive understanding of subsurface environments.



This initiative creates the first 3D geological model of a natural hydrogen resource drilled in Australia, enhancing understanding of H2 accumulation and migration. It supports H2 development while promoting engagement through an interactive web interface.

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2. Gold Hydrogen Limited. *AEP Perth Conference Presentation*. Available at: https://www.goldhydrogen.com.au/wp/wp-content/uploads/2024.05.23-AEP-Perth-Conference-Presentation.pdf (Accessed: 17 October 2024).
3. Gold Hydrogen Limited. *2024 Annual Report*. Brisbane, Australia: Gold Hydrogen Limited, 2024. Available at: https://www.goldhydrogen.com.au/wp/wp-content/uploads/202406_GHY_ANNUAL_REPORT.pdf (Accessed: 17 October 2024)
3. Department for Energy and Mining, the Government of South Australia, *South Australia Resources Information Gateway (SARIG)*. Available at: https://map.sarig.sa.gov.au/ (Accessed: 17 October 2024).